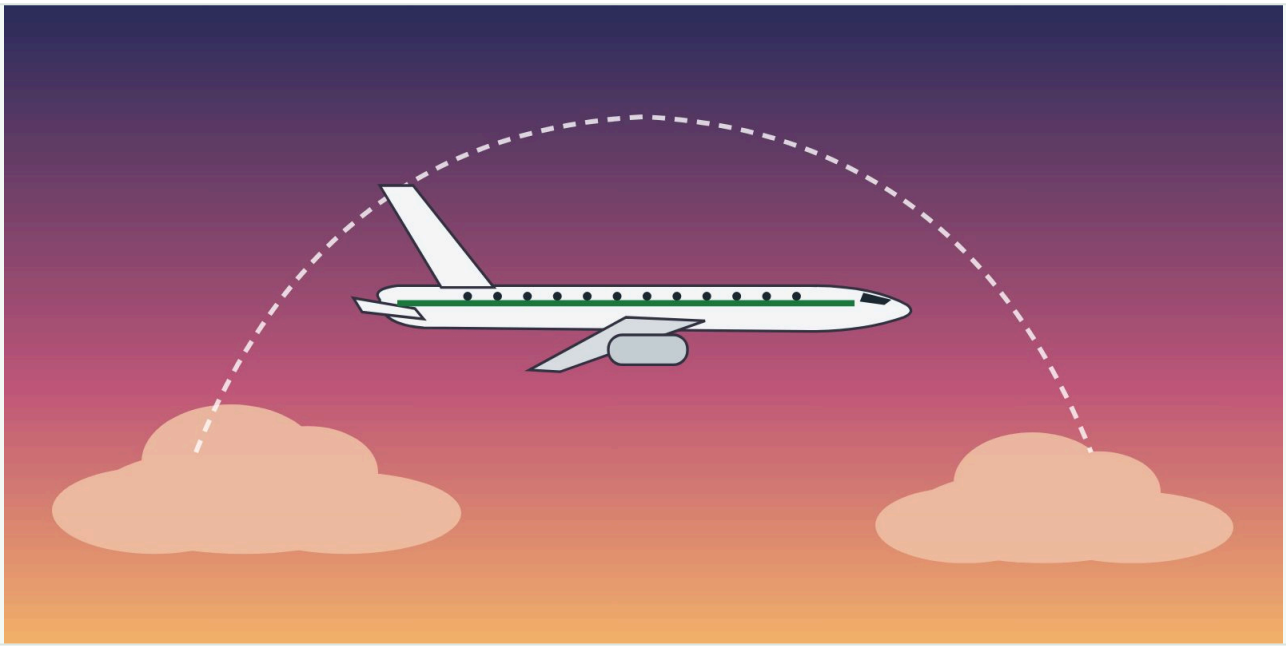
MODULE 3 · CERTIFICATION, PERFORMANCE AND LOADING

Lesson 09: Aeroplane Certification and the Flight Envelope

Lesson 9 of 24 · about 7 min

LEARNING OBJECTIVES

- Describe what certification does — and does not — demonstrate about aircraft handling.
- Identify the main speed and load limits of the flight envelope.
- Explain why upset recovery may take the aircraft outside the conditions tested in certification.



Certification demonstrates behaviour inside the normal flight envelope.

What certification demonstrates

Transport aeroplanes are certified to standards such as CS-25 and FAR Part 25. Test pilots demonstrate that, within the **normal flight envelope**, the aeroplane has adequate stability, controllability, stall warning and stall characteristics, and the required structural strength.

Certification does **not** demonstrate:

- recovery from every possible attitude — such as inverted flight or very high bank;
- behaviour far beyond the stall, at extreme sideslip, or combined with failures;
- how an average line pilot performs when startled at night.

Upsets can therefore take the aircraft into conditions where its behaviour is **less well known** — and where simulator data may also be limited (Lesson 24).

The flight envelope

Limit	Meaning

V_{MO} / M_{MO}	Maximum operating speed / Mach number — not to be deliberately exceeded
V_D / M_D	Design dive speed — demonstrated in flight test, with structural margin; not for line use
V_A	Design manoeuvring speed — full deflection of one control in one axis is acceptable below it (Lesson 6)
V_{FE}, V_{LE}, V_{LO}	Flap and landing gear speed limits
V_{S1g} / V_{SR}	Reference stall speed — the basis for take-off and approach speeds
Load factor limits	Typically +2.5 / −1.0 g clean, +2.0 / 0 g with flaps extended

Margins in normal operation

- Operating speeds include margins above the stall: for example, the final approach reference speed is typically about 1.23 × V_{SR} in the landing configuration.
- Manoeuvring speeds for each flap setting, and the speed bands on the PFD, show the safe range for normal manoeuvres.
- The buffet margin at cruise (Lesson 7) protects against turbulence and bank.

These margins are what prevention protects. Most developing upsets begin as an erosion of one of these margins — usually speed.

> **IN A RECOVERY**
> Exceeding a speed or load limit to save the aircraft is acceptable when there is no alternative — but avoid it if you can, and report any exceedance so that maintenance can inspect the aircraft.

KEY TAKEAWAYS

- Certification demonstrates behaviour within the normal envelope, not every possible upset.
- Know VMO/MMO, VA, flap/gear limits, stall reference speed and g limits.
- Operating speeds include margins; most upsets start with an eroded margin.
- Report any limit exceedance after a recovery.

KNOWLEDGE CHECK

Which statement about certification is correct?

A. Certification proves recovery from any attitude
B. Certification demonstrates handling within the normal flight envelope, not every upset condition
C. Certification tests are done with startled line pilots
D. VD may be used in normal line operations

Answer: B. Certification demonstrates stability, control and stall behaviour within the normal envelope. Extreme upset conditions may be outside what was tested.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.